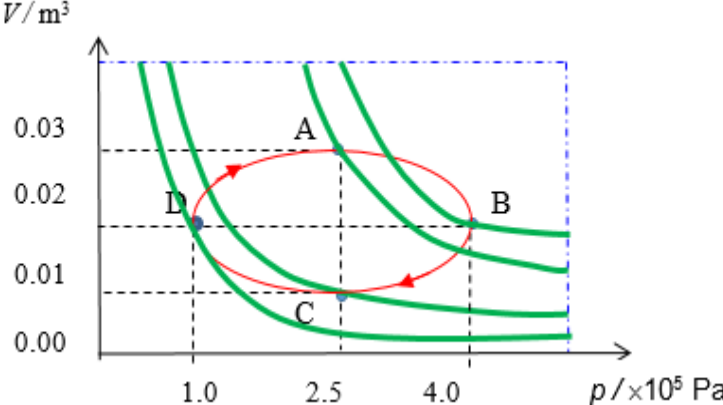


4	
(a)	Internal energy of an ideal gas is equal to the <u>random distribution</u> of <u>microscopic kinetic energy</u> OR <u>random distribution</u> of <u>molecular kinetic energy</u> OR <u>random distribution</u> of <u>kinetic energy of molecules</u>
(b)(i)	$T \propto pV$
	$pV = nRT$ $(2.5 \times 10^5) 0.03 = nRT_A = 7500$ $(4.0 \times 10^5) 0.02 = nRT_B = 8000$ $(2.5 \times 10^5) 0.01 = nRT_C = 2500$ $(1.0 \times 10^5) 0.02 = nRT_D = 2000$
	$T_D < T_C < T_A < T_B$

(b)(ii)																									
	All 4 isotherms drawn correctly.																								
(b)(iii)	After one complete cycle, the gas has the • <u>same temperature</u>, hence the • same random distribution of molecular kinetic energy, hence • same internal energy (no change in internal energy).																								
(b)(iv)	<table><tr><th>Process</th><th>$+\Delta U$</th><th>$+q$</th><th>$+w$</th></tr><tr><td>A $\rightarrow$ B</td><td>+</td><td>–</td><td>+</td></tr><tr><td>B $\rightarrow$ C</td><td>–</td><td>–</td><td>+</td></tr><tr><td>C $\rightarrow$ D</td><td>–</td><td>+</td><td>–</td></tr><tr><td>D $\rightarrow$ A</td><td>+</td><td>+</td><td>–</td></tr><tr><td>After one complete cycle</td><td>0</td><td>–</td><td>+</td></tr></table>	Process	$+\Delta U$	$+q$	$+w$	A $\rightarrow$ B	+	–	+	B $\rightarrow$ C	–	–	+	C $\rightarrow$ D	–	+	–	D $\rightarrow$ A	+	+	–	After one complete cycle	0	–	+
Process	$+\Delta U$	$+q$	$+w$																						
A $\rightarrow$ B	+	–	+																						
B $\rightarrow$ C	–	–	+																						
C $\rightarrow$ D	–	+	–																						
D $\rightarrow$ A	+	+	–																						
After one complete cycle	0	–	+																						
(b)(v)	$w = \left[\pi (0.01)(1.5 \times 10^5) \right]$																								
	$\Delta U = q + w$ $0 = q + \left[\pi (0.01)(1.5 \times 10^5) \right]$ $q = -4710 \text{ J}$																								
(c)	Yes. The expression is derived from																								
	1. <u>ideal gases</u> (hence $PV = NkT$)																								
	2. which obey the <u>Kinetic Theory of Gases</u> (hence $P = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$)																								

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